

§ 3.6.

P1. Solution:

(a) A - an 7×9 matrix, $r(A) = 5$.

$$\dim(C(A^T)) = 5$$

$$\dim(C(A)) = 5$$

$$\dim(N(A)) = 9 - 5 = 4 \rightarrow \text{sum} = 16$$

$$\dim(N(A^T)) = 7 - 5 = 2$$

(b) A - an 3×4 matrix, $r(A) = 3$

$$\dim(C(A)) = 3$$

$$\dim(N(A^T)) = 3 - 3 = 0$$

P4. Solution:

$$(a) \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$(b) \begin{bmatrix} 1 & -3 & 0 \\ 1 & -3 & 0 \\ 3 & -9 & 0 \end{bmatrix}$$

$$(c) \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$(d) \begin{bmatrix} 3 & 1 \\ -1 & -\frac{1}{3} \end{bmatrix}$$

$$(e) \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$$

P6. Solution:

$$(a) A = \begin{bmatrix} 0 & 3 & 3 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \xrightarrow[P_{23}]{P_{13}} \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 3 & 3 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = R$$

$\underbrace{\hspace{1.5cm}}_{\text{pivots}} \quad \underbrace{\hspace{1.5cm}}_{\text{free variables}}$

$$r(A) = 2$$

$$i) \dim(C(A^T)) = 2$$

$$\text{Basis: } \begin{bmatrix} 0 \\ 3 \\ 3 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$ii) \dim(C(A)) = 2$$

$$\text{Basis: } \begin{bmatrix} 3 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$iii) \dim(N(A)) = 4 - 2 = 2$$

$$\text{Basis: } \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 1 \\ 1 \end{bmatrix} \rightarrow \text{special solutions}$$

$$(iv) \dim(N(A^T)) = 3 - 2 = 1.$$

$$\text{solve } A^T y = 0$$

$$y_1 \begin{bmatrix} 0 \\ 3 \\ 3 \\ 3 \end{bmatrix} + y_2 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + y_3 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} = 0$$

$$\Rightarrow y_1 = y_3 = 0$$

$$\text{Basis: } \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$2. B = \begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = R, r(B) = 1.$$

$$i) \dim(C(B^T)) = 1$$

$$\text{Basis: } [1]$$

$$ii) \dim(C(B)) = 1$$

$$\text{Basis: } \begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix}$$

$$iii) \dim(N(A)) = 1 - 1 = 0$$

no Basis

$$ri) \dim(N(A^T)) = 3 - 1 = 2$$

$$\text{solve } B^T y = 0$$

$$y_1 + 4y_2 + 5y_3 = 0$$

$$\Rightarrow s_1 = \begin{bmatrix} -4 \\ 1 \\ 0 \end{bmatrix}, s_2 = \begin{bmatrix} -5 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{Basis: } \begin{bmatrix} -4 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -5 \\ 0 \\ 1 \end{bmatrix}.$$

P.9. Solution:

$$(a) [A] \text{ and } \begin{bmatrix} A \\ A \end{bmatrix}$$

The row space and nullspace are the same.

$$(b) \begin{bmatrix} A \\ A \end{bmatrix} \text{ and } \begin{bmatrix} A & A \\ A & A \end{bmatrix}$$

The column space and left nullspace are the same.

P12. Solution:

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

This can't be a basis for the row space and nullspace at the same time, since the row space is always perpendicular to the nullspace.

P26. solution: B B A.

S4.1.

P3. Solution:

(a) $\begin{bmatrix} 1 & 2 & -3 \\ 2 & -3 & 1 \\ -3 & 5 & -2 \end{bmatrix}$

(b) impossible.
 $2-3+5 \neq 0$

(c) impossible.

$$Ax = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$A^T \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow [1 \ 0 \ 0]A = [0 \ 0 \ 0]$$

$$\Rightarrow [1 \ 0 \ 0]Ax = [0 \ 0 \ 0] \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$[1 \ 0 \ 0] \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 0$$

$1 = 0 \rightarrow \text{Contradiction}$

(d) impossible.

$$A^T A = 0 \Rightarrow A = 0$$

(e) impossible.

$$(a_{ij}) = A = [a_1 \ a_2 \ \dots \ a_n] = \begin{bmatrix} a^1 \\ a^2 \\ \vdots \\ a^m \end{bmatrix}$$

$$\sum_{i=1}^n a_i = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \sum_{i=1}^n \sum_{j=1}^m a_{ij} = 0$$

$$\sum_{j=1}^m a_j = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \Rightarrow \sum_{j=1}^m \sum_{i=1}^n a_{ij} = n > \text{Contradiction}$$

P6. Solution:

$$A \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \\ 9 \end{bmatrix} \text{ where } A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 2 & 3 \\ 3 & 4 & 5 \end{bmatrix}$$

$$\begin{cases} y_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + y_2 \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix} + y_3 \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix} = 0 \Rightarrow A^T y = 0 \\ 5y_1 + 5y_2 + 9y_3 = 1 \Rightarrow y^T b = 1 \end{cases}$$

$$A^T = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & -2 & -2 \\ 0 & -1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & -2 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & -2 & -2 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} = R$$

$$Ry = 0 \Rightarrow s_1 = \begin{bmatrix} -1 \\ -1 \end{bmatrix} \Rightarrow y = c \begin{bmatrix} -1 \\ -1 \end{bmatrix} = \begin{bmatrix} -c \\ -c \end{bmatrix}$$

$$5(-c) + 5(-c) + 9c = 1 \\ \Rightarrow -c = 1 \Rightarrow c = -1 \\ \Rightarrow y = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$y \in N(A^T)!$$

P7. Solution:

$$\begin{aligned} x_1 - x_2 &= 1 \\ x_2 - x_3 &= 1 \\ x_1 - x_3 &= 1 \end{aligned} \Rightarrow \begin{aligned} \frac{1}{3}(x_1 - x_2) + \frac{1}{3}(x_2 - x_3) \\ + \frac{1}{3}(x_1 - x_3) &= \frac{1}{3} + \frac{1}{3} + \frac{1}{3} \\ \Rightarrow 0 &= 1. \end{aligned}$$

P9. Solution:

Column space

orthogonal to the nullspace of A^T

P12. Solution:

$$A = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \Rightarrow r(A) = 1$$

$$\dim(C(A^T)) = 1$$

$$\text{Basis: } \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\dim(N(A)) = 2 - 1 = 1$$

$$\text{Basis: } \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

These two vectors $\begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ form a basis for $\mathbb{R}^2$. $x_r = x_n$

$$x = \begin{bmatrix} 2 \\ 0 \end{bmatrix} = a \begin{bmatrix} 1 \\ -1 \end{bmatrix} + b \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\Rightarrow a = 1, b = 1.$$

$$\Rightarrow x = x_r + x_n \text{ where } x_r = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, x_n = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

P16. Proof:

$$y \in N(A^T) \text{ means } A^T y = 0$$

$$\Rightarrow y^T (Ax) = y^T A x = (y^T A^T y)^T = 0$$

$$\Rightarrow y \perp Ax.$$

P18. Solution:

$$A = \begin{bmatrix} 1 & 5 & 1 \\ 2 & 2 & 2 \end{bmatrix}.$$